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CST-250 Programming in C# II
Grand Canyon University
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Milestone 2

Files

<https://github.com/CGalloway3/CST-250-Projects/tree/master/Milestone2>

Video

<https://www.loom.com/share/cb92311abbfa489fbc2b720fe8b8a580>

UML Class Diagram

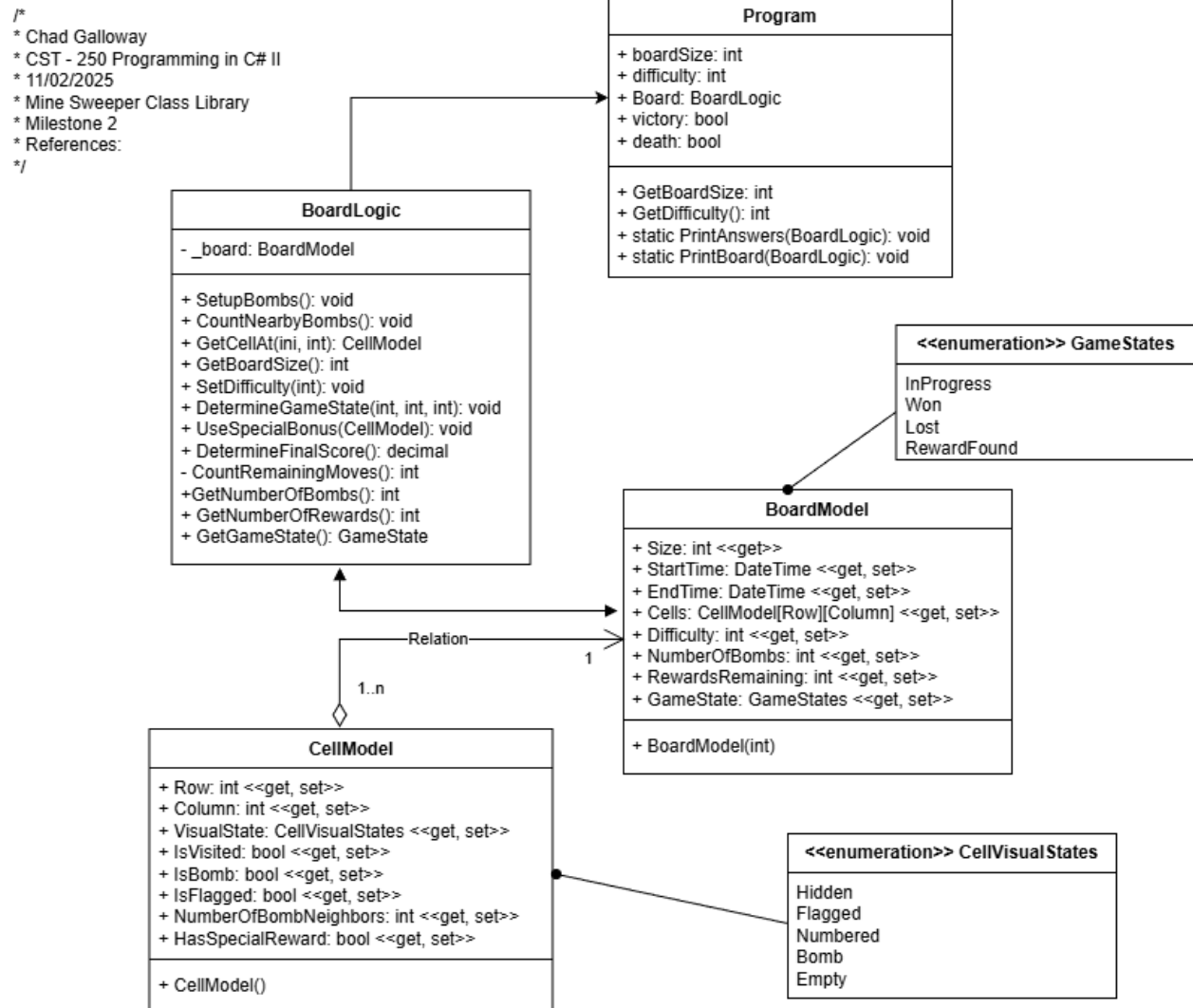
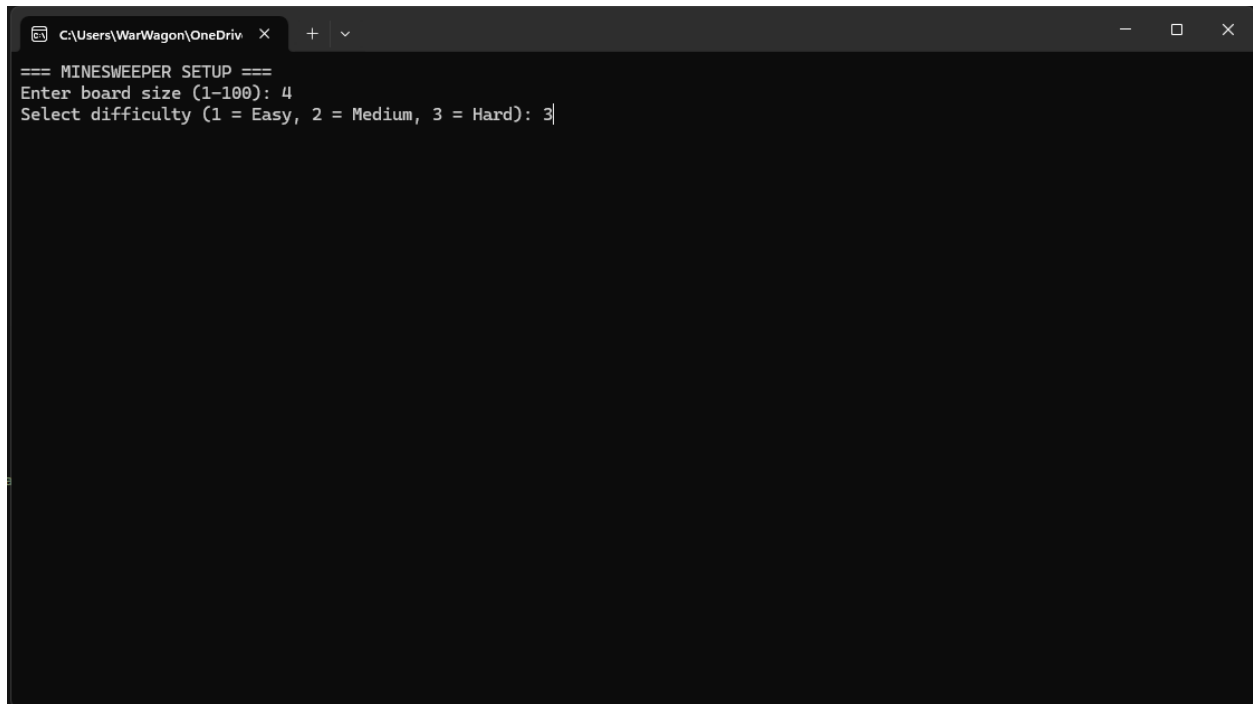


Figure 1: UML Class Diagram for the Milestone 2 changes

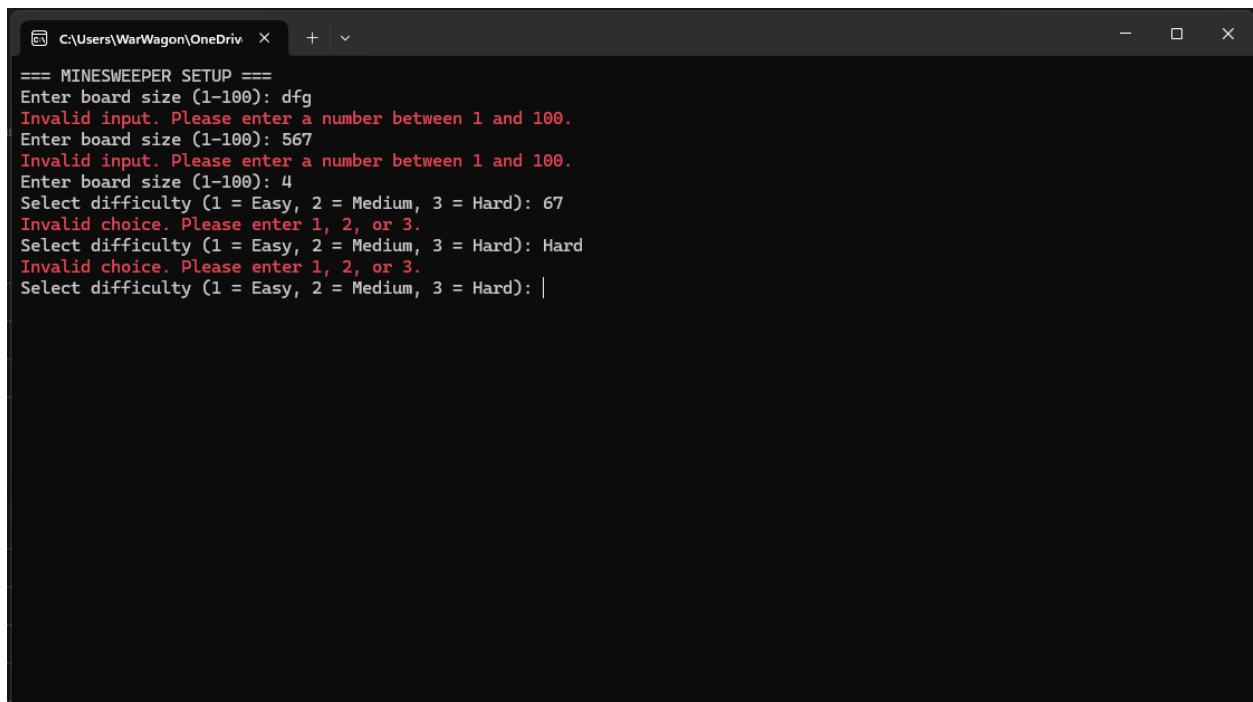
Screen Shots



A screenshot of a Windows command prompt window titled "C:\Users\WarWagon\OneDrive". The window shows the following text:

```
=== MINESWEEPER SETUP ===  
Enter board size (1-100): 4  
Select difficulty (1 = Easy, 2 = Medium, 3 = Hard): 3|
```

Figure 2: Screenshot of user setup for the console app



A screenshot of a Windows command prompt window titled "C:\Users\WarWagon\OneDrive". The window shows the following text:

```
=== MINESWEEPER SETUP ===  
Enter board size (1-100): dfg  
Invalid input. Please enter a number between 1 and 100.  
Enter board size (1-100): 567  
Invalid input. Please enter a number between 1 and 100.  
Enter board size (1-100): 4  
Select difficulty (1 = Easy, 2 = Medium, 3 = Hard): 67  
Invalid choice. Please enter 1, 2, or 3.  
Select difficulty (1 = Easy, 2 = Medium, 3 = Hard): Hard  
Invalid choice. Please enter 1, 2, or 3.  
Select difficulty (1 = Easy, 2 = Medium, 3 = Hard): |
```

Figure 3: Screenshot of app handling errors

```
C:\Users\WarWagon\OneDrive  X  +  v

Welcome to Mine Sweeper Console Application!

Board size: 4x4, Difficulty: Medium
Good Luck!!

Here is the answer key for the board.
  0  1  2  3
+---+---+---+---+
0 | . | 1 | B | 1 |
+---+---+---+---+
1 | . | 1 | 1 | 1 |
+---+---+---+---+
2 | . | 1 | r | 1 |
+---+---+---+---+
3 | . | 1 | B | 1 |
+---+---+---+---+

Here is the current board.
  0  1  2  3
+---+---+---+---+
0 | ? | ? | ? | ? |
+---+---+---+---+
1 | ? | ? | ? | ? |
+---+---+---+---+
2 | ? | ? | ? | ? |
+---+---+---+---+
3 | ? | ? | ? | ? |
+---+---+---+---+

There are still 2 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3):
```

Figure 4: Screenshot of Game Starting

```
C:\Users\WarWagon\OneDrive  X  +  v

  0  1  2  3
+---+---+---+---+
0 | ? | ? | ? | ? |
+---+---+---+---+
1 | ? | ? | ? | ? |
+---+---+---+---+
2 | ? | ? | ? | ? |
+---+---+---+---+
3 | ? | ? | ? | ? |
+---+---+---+---+

There are still 2 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3): 2
Enter column (0 - 3): 2
Enter choice ( 1 = Visit Cell, 2 = Flag Cell ): 1
Here is the current board.
  0  1  2  3
+---+---+---+---+
0 | ? | ? | ? | ? |
+---+---+---+---+
1 | ? | ? | ? | ? |
+---+---+---+---+
2 | ? | ? | 1 | ? |
+---+---+---+---+
3 | ? | ? | ? | ? |
+---+---+---+---+

*** You found a reward ***
There are still 2 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3): |
```

Figure 5: Screenshot of User earning reward

```
C:\Users\WarWagon\OneDrive X + v
0 | ? | ? | ? | ? |
+---+
1 | ? | ? | ? | ? |
+---+
2 | ? | ? | 1 | ? |
+---+
3 | ? | ? | ? | ? |
+---+
*** You found a reward ***
There are still 2 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3): 0
Enter column (0 - 3): 2
Enter choice ( 1 = Visit Cell, 2 = Flag Cell, 3 = Use Reward (Bomb Detector) ): 3
That cell does have a bomb
Here is the current board.
  0  1  2  3
+---+
0 | ? | ? | ? | ? |
+---+
1 | ? | ? | ? | ? |
+---+
2 | ? | ? | 1 | ? |
+---+
3 | ? | ? | ? | ? |
+---+
Game in Progress:
There are still 2 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3):
```

Figure 6: Screenshot of user using up his reward on a square

```
C:\Users\WarWagon\OneDrive X + v
+---+
0 | ? | ? | ? | ? |
+---+
1 | ? | ? | ? | ? |
+---+
2 | 2 | 2 | 2 | . |
+---+
3 | ? | ? | ? | ? |
+---+
Game in Progress:
There are still 2 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3): 1
Enter column (0 - 3): 1
Enter choice ( 1 = Visit Cell, 2 = Flag Cell, 3 = Use Reward (Bomb Detector) ): 2
Here is the current board.
  0  1  2  3
+---+
0 | ? | ? | ? | ? |
+---+
1 | ? | F | ? | ? |
+---+
2 | 2 | 2 | 2 | . |
+---+
3 | ? | ? | ? | ? |
+---+
Game in Progress:
There are still 1 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3):
```

Figure 7: Screenshot of user flagging a bomb

```
C:\Users\WarWagon\OneDrive X + v
+---+---+---+---+
0 | ? | ? | ? | ? |
+---+---+---+---+
1 | ? | F | ? | ? |
+---+---+---+---+
2 | 2 | 2 | 2 | . |
+---+---+---+---+
3 | ? | ? | ? | ? |
+---+---+---+---+
Game in Progress:
There are still 1 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3): 1
Enter column (0 - 3): 1
Enter choice ( 1 = Visit Cell, 2 = Flag Cell, 3 = Use Reward (Bomb Detector) ): 2
Here is the current board.
    0 1 2 3
+---+---+---+---+
0 | ? | ? | ? | ? |
+---+---+---+---+
1 | ? | ? | ? | ? |
+---+---+---+---+
2 | 2 | 2 | 2 | . |
+---+---+---+---+
3 | ? | ? | ? | ? |
+---+---+---+---+
Game in Progress:
There are still 2 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3): |
```

Figure 8: Screenshot of Unflagging a cell

```
Microsoft Visual Studio Debu X + v
1 | . | 1 | 1 | 1 |
+---+---+---+---+
2 | . | 1 | 1 | 1 |
+---+---+---+---+
3 | . | 1 | F | ? |
+---+---+---+---+
Game in Progress:
There are still 0 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3): 3
Enter column (0 - 3): 3
Enter choice ( 1 = Visit Cell, 2 = Flag Cell ): 1
Here is the current board.
    0 1 2 3
+---+---+---+---+
0 | . | 1 | F | 1 |
+---+---+---+---+
1 | . | 1 | 1 | 1 |
+---+---+---+---+
2 | . | 1 | 1 | 1 |
+---+---+---+---+
3 | . | 1 | F | 1 |
+---+---+---+---+
Congratulations you have beaten the game. Well Done.

C:\Users\WarWagon\OneDrive\School\Classes\CST-250\repo\Milestone2\MineSweeperClassLibrary\MineSweeperConsoleApp\bin\Debu
g\net9.0\MineSweeperConsoleApp.exe (process 32536) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .|
```

Figure 9: Screenshot of Winning the game

```
Microsoft Visual Studio Debu X + v
3 | ? | ? | ? | ? |
+---+---+---+---+
Game in Progress:
There are still 2 bombs that are unaccounted for.
What is your next move.
Enter row (0 - 3): 1
Enter column (0 - 3): 1
Enter choice ( 1 = Visit Cell, 2 = Flag Cell, 3 = Use Reward (Bomb Detector) ): 1
Here is the current board.
  0  1  2  3
+---+---+---+---+
0 | ? | ? | ? | ? |
+---+---+---+---+
1 | ? | B | ? | ? |
+---+---+---+---+
2 | 2 | 2 | 2 | . |
+---+---+---+---+
3 | ? | ? | ? | ? |
+---+---+---+---+
BOOM!!!
      BOOM!!!
      BOOM!!!
You found a bomb.
Awe, better luck next time, you almost had it.

C:\Users\WarWagon\OneDrive\School\Classes\CST-250\repo\Milestone2\MineSweeperClassLibrary\MineSweeperConsoleApp\bin\Debug\net9.0\MineSweeperConsoleApp.exe (process 54756) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .|
```

Figure 10: Screenshot of Game Loss

Summary of Key Concepts

Some of the key concepts practiced this week were Abstraction, Inheritance, and Polymorphism. Abstraction was practiced by hiding elements of the board model behind a private `_board` variable. This hides things about the `_board` we don't want public to the app, like the number of rewards. This stops a malicious app from setting the number of rewards available to 100 and cheating the game. The only way to interact with the board if you are not board logic is through the accessors exposed by the board logic like `GetNumberOfRewards()`. This allows another class to find out how many rewards exist but board logic does not expose any way to increase the number of rewards. The earning of rewards has to happen naturally in the execution of other methods in the board logic. Inheritance is everywhere in C# and it involves a class being formed out of the basic structure of another class. There aren't many really good examples of this in the small minesweeper app but they exist in almost every C# class they all inherit from some base class. Lastly is polymorphism. This refers to one class not being concerned with how another class uses its methods. For example, the `program.cs` class doesn't really care how the use special bonus happens. It just tells the `boardlogic` class hey I want to use a reward and the logic takes care of the rest. This is particularly evident when console apps and gui apps share the same backend, They are setup to not be concerned with the process of the what the backend has to do and just concentrate on the results.